KTrade v10.3

End-to-End Project Document

Project folder: C:\trading-agent\Ktrade_v10_3

A complete plain-English guide covering project workflow, strategy, market data, VectorBT, sentiment, dashboard, Alpaca paper trading, and daily commands.

Area	Covered In This Document
Strategy	Momentum, volume, trend, MACD, relative strength, ORB
Backtest	VectorBT approved parameters and intraday testing
Sentiment	Finnhub news + FinBERT score + OpenAI explanation
Trading	Alpaca paper orders and broker reconciliation
Operations	Daily commands, testing, deployment notes

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This document explains how the KTrade v10.3 paper-trading project works end to end. It covers the dashboard, backend, market data, strategy engine, VectorBT backtesting, sentiment layer, broker reconciliation, risk rules, and daily run process.

Important note: This project is for paper trading, testing, and learning. It is not financial advice. Do not use it for live-money trading unless the strategy has been independently validated and risk controls are reviewed.

1. What This Project Does

KTrade v10.3 is a paper-trading dashboard and signal engine.

The goal is:

1. Collect market data.
2. Run strategy logic on selected stocks/ETFs/crypto symbols.
3. Rank symbols as BUY or WATCH.
4. Show signal details in a dashboard.
5. Add AI sentiment confirmation using Finnhub + FinBERT + OpenAI.
6. Allow paper trades through Alpaca.
7. Reconcile dashboard positions with real Alpaca paper broker truth.

Simple flow:

```
Market Data -> Strategy Scanner -> BUY/WATCH Signals -> Dashboard -> Paper Trade -> Alpaca Broker Truth
-> Sentiment Layer -> Confirmation / Warning
```

2. Main Components

```
| Component | What It Does |
|---|---|
| Frontend Dashboard | Shows signals, positions, trades, alerts, performance, and AI sentiment. |
| Backend API | Serves dashboard data and connects to Alpaca, Finnhub, reconciliation, and sentiment. |
| Scanner Agent | Reads market data and ranks symbols using strategy/conviction score. |
| VectorBT Backtest | Tests strategies historically and saves approved parameters. |
| Intraday VectorBT | Tests intraday strategy ideas on shorter timeframes. |
| Sentiment Engine | Uses Finnhub news, FinBERT sentiment, and OpenAI explanation. |
| Alpaca Paper Trading | Places paper trades and returns real paper account positions/orders. |
| Broker Reconciliation | Checks that dashboard/account state matches Alpaca broker truth. |
```

3. Current Important Settings

Current project settings from your `.env`:

```
LIVE_TRADING=false KTRADE_PAPER_ORDER_SUBMISSION=true RISK_PER_TRADE=0.005 MIN_CONVICTION_SCORE=75
MAX_POSITIONS_PER_HEARTBEAT=3 KTRADE_SCAN_UNIVERSE=extended KTRADE_ENABLE_OPENAI_SENTIMENT=true
KTRADE_ENABLE_FINBERT=true KTRADE_SENTIMENT_TTL_SECONDS=1800
```

Meaning:

- Live trading is disabled.

• Paper trading is enabled.
• Each trade is risk-limited.
• A signal needs at least 75 conviction to become BUY.
• Below 75, it should normally show WATCH.
• Sentiment is enabled as a confirmation/warning layer.

4. Data Sources

The project supports multiple market data sources.

```
| Source | Use |
|---|---|
| Polygon | Market bars/quotes when available. Good primary source if API is stable. |
| Alpaca | Broker account, paper orders, positions, account equity, and quote fallback. |
| Finnhub | News, quote/candle fallback, sentiment news source. |
| yfinance | Final fallback for historical data/testing. |
```

Recommended data order:

```
Polygon -> Alpaca -> Finnhub -> yfinance
```

If you force Finnhub scanner:

```
set KTRADE_DATA_PROVIDER=finnhub
```

Then the scanner should show:

```
DataFeed source: FINNHUB
```

5. What Is Our Strategy-

Your project does not use one single simple strategy. It uses a combined conviction strategy.

The scanner looks at several market conditions and creates a conviction score.

Daily Scanner Strategy Inputs

For daily/swing-style scan, the score uses:

```
| Component | Meaning |
|---|---|
| Momentum | Checks whether the stock has moved strongly over recent periods. |
| Volume | Checks whether volume is stronger than normal. |
| Trend | Checks whether price is above key moving averages like 50 EMA and 200 EMA. |
| MACD | Checks whether MACD supports bullish or bearish movement. |
| Relative Strength | Checks whether the stock is stronger than SPY. |
```

Daily weighting from the scanner logic:

```
Momentum: 33% Volume: 22% Trend: 23% MACD: 16% Relative Strength: remaining small confirmation ORB:
excluded on daily because opening range needs intraday candles
```

Intraday Scanner Strategy Inputs

For intraday scan, ORB is included.

Intraday weighting:

```
Momentum: 30% Volume: 20% Trend: 20% MACD: 15% ORB: 10% Relative Strength: 5%
```

BUY / WATCH Rule

Current threshold:

```
MIN_CONVICTION_SCORE=75
```

Meaning:

```
Score >= 75 -> BUY Score < 75 -> WATCH
```

Example:

```
NVDA score 82 -> BUY AAPL score 68 -> WATCH
```

6. Strategy Types Used

6.1 Momentum Strategy

Momentum checks whether price has been moving strongly.

Simple meaning:

```
If a stock is already moving up strongly and the move has strength, it can score higher.
```

Used for:

• Trend continuation

• Strong movers

• Growth/AI/tech names

6.2 MACD + EMA Strategy

MACD checks momentum direction. EMA checks trend.

BUY support happens when:

```
Price is above long-term trend MACD turns bullish Trend condition is supportive
```

Simple example:

```
Price above 200 EMA + MACD crosses up = bullish confirmation
```

6.3 MACD EMA Confluence

This is a stronger version of MACD + EMA.

It adds:

• Volume surge

• Higher-high confirmation

• Trend confirmation

Meaning:

```
The stock is not just moving up; it has trend, volume, and price confirmation.
```

6.4 ORB - Opening Range Breakout

ORB is for intraday trading.

Simple meaning:

```
First part of the trading day creates a high and low range. If price breaks above that range with volume,
it may be an intraday BUY setup.
```

Important:

• ORB should be used only with intraday candles like 1m, 5m, or 15m.

• It is not meaningful for daily candles.

• v10.3 correctly excludes ORB from daily scoring.

6.5 Relative Strength

Relative strength compares a stock against SPY.

Simple meaning:

```
If the stock is stronger than the broader market, it gets extra confirmation.
```

7. VectorBT Backtesting Role

VectorBT is not the daily scanner. It is the historical testing engine.

It checks whether strategy parameters worked historically.

Flow:

```
Historical data -> VectorBT tests strategy parameters -> Approved params saved -> Scanner can use
approved params
```

Main files:

```
ktrade_vectorbt.py ktrade_intraday_vectorbt.py data\ktrade_approved_params.json
data\ktrade_intraday_approved_params.json
```

VectorBT tests strategies like:

• MACD

• EMA

• Momentum

• Conviction strategy

• Intraday VWAP/ORB-style ideas in the intraday file

It checks metrics such as:

```
| Metric | Meaning |
|---|---|
| Sharpe | Risk-adjusted quality of returns. Higher is better. |
| Profit Factor | Gross profit divided by gross loss. Above 1 is better. |
| Max Drawdown | Largest historical decline. Lower is safer. |
| Win Rate | Percent of winning trades. Useful but not enough alone. |
| Total Return | Historical return in the test period. |
```

Important:

```
VectorBT approval does not guarantee future profit. It only says the setup passed historical rules.
```

8. Sentiment Layer - Finnhub + FinBERT + OpenAI

Sentiment is now added as a confirmation layer.

Flow:

```
Finnhub News -> FinBERT Sentiment Score -> OpenAI Explanation -> Dashboard Confirmation
```

Finnhub Role

Finnhub gets company news headlines and summaries.

Example:

```
/news/NVDA
```

FinBERT Role

FinBERT is a finance-specific AI model.

It reads financial news and scores:

```
Positive / Neutral / Negative Confidence score
```

OpenAI Role

OpenAI explains the result in simple English.

Example:

```
Recent news is positive because analysts mention strong AI demand and revenue growth. This supports the
existing BUY signal, but strategy and risk rules should still decide.
```

Sentiment Rule

Sentiment must not place trades by itself.

Correct use:

```
Strategy BUY + Sentiment Positive = BUY confirmed Strategy BUY + Sentiment Negative = BUY with warning
Strategy WATCH + Sentiment Positive = Watch improving Strategy WATCH + Sentiment Negative = Risk warning
```

9. Paper Trading Flow

Paper trading goes through Alpaca.

Flow:

```
Dashboard Trade Button -> Backend /buy or /sell -> Alpaca Paper API -> Order/Position update -> Dashboard
sync
```

Important:

```
LIVE_TRADING=false
```

So the project is intended for paper trading.

Paper order submission is enabled:

```
KTRADE_PAPER_ORDER_SUBMISSION=true
```

This means dashboard paper orders are allowed.

10. Broker Reconciliation

Broker reconciliation checks whether your dashboard agrees with Alpaca.

Why it matters:

• The dashboard should not invent positions.

• Alpaca is the source of truth.

• If a trade fills, cancels, or closes, dashboard should reflect broker reality.

Main endpoints:

```
/reconciliation /reconcile_truth /closed_trades
```

Simple meaning:

```
If Alpaca says you hold 3 positions, dashboard should show the same 3 positions.
```

11. Dashboard Explanation

Main dashboard URL:

```
http://localhost:5001
```

Main tabs:

Tab	Purpose
Signals	Shows BUY/WATCH signals from scanner.
Positions	Shows current Alpaca paper positions.
Trade	Lets you place paper trades.
Alerts	Shows warnings and system messages.
Performance	Shows account/trade performance.
AI Advisor	Chat/assistant-style project notes if enabled.

Signal card fields:

Field	Meaning
Ticker	Stock symbol.
CALL/PUT	Direction/type display for paper option-style signal cards.
Strategy	Strategy that produced the signal, like MOMENTUM or MACD_EMA.
BUY/WATCH	Scanner decision based on conviction score.
Entry	Approximate entry price shown by system.
Target	Expected target level.
Stop	Stop/risk level.
Live Est	Current estimated value from latest price.
Conviction	Score from strategy engine.
AI Sentiment	FinBERT/OpenAI confirmation or warning.

12. Main Files and Uses

File	Use
backend\ktrade_alpaca.py	Main backend server, dashboard API, Alpaca connection, paper orders, reconciliation, sentiment endpoint.
frontend\KTrade_preview.html	Main dashboard UI shown at localhost:5001.
agent\ktrade_agent_v9.py	Scanner and strategy engine. Produces BUY/WATCH signals.
data\ktrade_data.py	Market data provider logic: Polygon, Alpaca, Finnhub, yfinance.
data\openai_finbert_sentiment.py	Sentiment engine: Finnhub news + FinBERT + OpenAI explanation.
backend\broker_reconciler.py	Broker truth and closed trade reconciliation.
ktrade_vectorbt.py	Daily/swing historical backtesting.
ktrade_intraday_vectorbt.py	Intraday historical backtesting.
data\ktrade_scan_latest.json	Latest scanner output used by dashboard.
data\ktrade_approved_params.json	VectorBT-approved daily strategy parameters.
data\ktrade_intraday_approved_params.json	VectorBT-approved intraday strategy parameters.
.env	Local secret/config file. Do not upload to GitHub.
requirements.txt	Python dependencies.
RUN_SCANNER_FINNHUB_DAILY.cmd	Runs daily scanner using Finnhub.
RUN_SCANNER_FINNHUB_INTRADAY.cmd	Runs intraday scanner using Finnhub.
START_BACKEND.cmd	Starts backend if present/updated.

13. Daily Run Commands

CMD 1 - Start Backend

```
cd /d C:\trading-agent\Ktrade_v10_3 .venv\Scripts\python.exe backend\ktrade_alpaca.py
```

Keep this window open.

Dashboard:

```
http://localhost:5001
```

CMD 2 - Run Daily Scanner

```
cd /d C:\trading-agent\Ktrade_v10_3 RUN_SCANNER_FINNHUB_DAILY.cmd
```

Expected source:

```
DataFeed source: FINNHUB
```

CMD 2 - Run Intraday Scanner

```
cd /d C:\trading-agent\Ktrade_v10_3 RUN_SCANNER_FINNHUB_INTRADAY.cmd
```

Use this when you want intraday-style signals.

14. VectorBT Commands

Run daily VectorBT only when:

• Strategy changed
• New stocks added
• You want fresh approved parameters
• You want to recheck historical performance

```
cd /d C:\trading-agent\Ktrade_v10_3 .venv\Scripts\python.exe ktrade_vectorbt.py
```

Run intraday VectorBT:

```
cd /d C:\trading-agent\Ktrade_v10_3 .venv\Scripts\python.exe ktrade_intraday_vectorbt.py --fast
--universe extended
```

You do not need to run VectorBT every time you open the dashboard.

15. Sentiment Test Commands

Start backend first, then open:

```
http://localhost:5001/sentiment/AAPL
```

Good result should include:

```
"available": true "sentiment": "positive" or "neutral" or "negative" "finbert_used": true "openai_used":
true
```

If FinBERT is not installed, it may show:

```
"finbert_used": false
```

Then install:

```
cd /d C:\trading-agent\Ktrade_v10_3 .venv\Scripts\pip.exe install transformers torch
```

If OpenAI key is missing or invalid, it may show:

```
"openai_used": false
```

Then check:

```
C:\trading-agent\Ktrade_v10_3\.env
```

16. How to Read a BUY Signal

Example:

```
Ticker: NVDA Signal: BUY Score: 82 Strategy: MOMENTUM Sentiment: Positive
```

Meaning:

1. Scanner score is above 75.
2. Strategy conditions are strong enough.
3. Sentiment confirms the signal.
4. It may be considered for paper trade.
5. Risk rules still matter.

This does not mean guaranteed profit.

17. How to Read a WATCH Signal

Example:

```
Ticker: AAPL Signal: WATCH Score: 68 Strategy: MOMENTUM Sentiment: Positive
```

Meaning:

1. Some conditions are good.
2. Score is below 75.
3. It is not ready as a BUY under your current rules.
4. Sentiment may be improving, but strategy still says WAIT.

18. Why a Position Can Be Red While Signal Still Says BUY

A stock can be temporarily down after entry while the strategy still says BUY.

Reasons:

• Price moved against you after entry.
• Signal is based on broader trend, not every small price move.
• Spread/market movement can create temporary paper loss.
• Strategy can remain BUY until conditions break.

A red position does not automatically mean strategy failed.

A strategy failure is better judged by:

• Did stop-loss hit-
• Did risk rule trigger-
• Did thesis/strategy condition break-
• Did many trades fail over time-
• Does backtest/live paper performance remain positive-

19. One-Day / Intraday vs Swing

```
| Type | Meaning |
|---|---|
| Intraday | Trade opened and usually closed same day. Uses 1m/5m/15m signals. |
| Swing | Trade held for days/weeks. Uses daily trend and momentum. |
| Momentum | Follows strong price movement. Can be intraday or swing. |
| ORB | Opening Range Breakout, mostly intraday. |
| Position/Invest | Longer-term holding. Not the current primary project goal. |
```

20. Production / Render Notes

For a public link, backend must run on a server like Render, DigitalOcean, Azure, etc.

Render Web Service:

• Runs backend/dashboard.
• Must have environment variables configured.
• Users can open the public Render URL.

Render Cron Job:

• Can run scanner periodically.

• Example: daily scanner every few minutes/hours during market time.

Important:

• If backend is not running, dashboard cannot update.

• If scanner is not running, signals may become stale.

• If Alpaca/Finnhub/OpenAI keys are missing, related features will not work.

21. Safety Rules

Recommended safety rules:

1. Keep LIVE_TRADING=false until testing is complete.
2. Use paper trading only.
3. Never commit `.env` to GitHub.
4. Rotate API keys if they appear in screenshots/logs.
5. Keep sentiment as confirmation only.
6. Do not average into losing trades unless a tested rule explicitly allows it.
7. Watch max drawdown and profit factor over many trades, not one trade.
8. Do not trust one backtest blindly.
9. Track real paper results for weeks/months.

22. Best Daily Workflow

1. Start backend.
2. Run scanner.
3. Open dashboard.
4. Review BUY signals.
5. Click signal card and check:

• Strategy

• Score

• Entry

• Target

• Stop

• Sentiment

• Risk/reward

6. Place only paper trade if you want to test.
7. Monitor Positions tab.
8. Sync/refresh after trade.
9. Review closed trades and performance.
10. Adjust strategy only after enough trade sample size.

23. Final Simple Explanation

KTrade v10.3 is built like this:

```
VectorBT checks if strategy worked historically. Scanner checks current market conditions. Finnhub gets
news. FinBERT scores financial sentiment. OpenAI explains the sentiment. Dashboard shows BUY/WATCH with
reasoning. Alpaca places paper trades. Broker reconciliation checks the truth. Risk rules protect the
```

```
account.
```

The main decision should always be:

```
Strategy + Risk + Backtest + Paper Results
```

Sentiment should only be:

```
Confirmation or warning
```

Current project status:

```
Paper trading: enabled Live trading: disabled BUY threshold: 75 Sentiment: Finnhub + FinBERT + OpenAI
available when keys/packages are configured
```